

Xtreme Adventures Tours

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IS 421: Systems Analysis and Design

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Introduction

Project Overview:

The Xtreme automated booking system (XABS), a brand-new automated booking system, will provide automated processes, real-time updates, and elevate user experience to improve operational efficiency, system reliability, customer satisfaction, and business growth. The purpose of this project is to replace the existing error-prone manual system with a revolutionary solution to improve the customer experience and make a business successful.

Recommendation:

We recommend that creating a customized software solution due to these reasons:

- A customized software solution is made to meet some specific requirements to fit Xtreme Adventures Tours' needs, such as removing the existing manual processes with automated functionality, real-time availability system, etc.
- Enhance scalability to allow the system handling company's growth.
- Seamless integration with exist systems and database to improve overall efficiency
- Offering unique features that cannot be found in off-the-shelf solution.

Baseline Project Plan

System Description (V&G Page 127 – 128)

Alternatives:

Outsourcing development:

- Pros: Quicker implementation, lower cost, regular updates and maintenance handled by vendor.
- Challenges: Limited customization, subscription fees, and potential integration challenges.

In-house development:

- Pros: Full control, customization, and alignment with business goals.
- Challenges: Resource intensive (Cost & time), and potential skill gaps.

System Description:

The system will provide a comprehensive user interface to allow users to explore, search, plan, and book effectively, in addition, staff in the company to learn and adapt efficiently. The backend part of the system will be fully automated for checking availability, booking management, secure handling payment procedures, then sending confirmation information to customers also to company to receive the notification.

Feasibility Assessment (V&G Page 129)

Economic Analysis (V&G Page 113 – 120)

XABS will potentially drive business growth and increase customer retention through the improvement of service reliability. Moreover, the automated system can significantly reduce labor cost.

Technical Analysis (V&G Page 121 – 124)

XABS being developed to have a reliable automated system, therefore, its requirements involve reliable real-time services, solid cloud hosting, secure database, safe payment methods, and robust scalability.

Operational Analysis (V&G Page 124)

The new automated systems will significantly improve Xtreme Adventure Tours' workflows due to the system's efficiency and error reduction. Employees will have access to precise real-time tour information, and straightforward procedures to manage bookings or cooperate with both business partners and customers seamlessly.

Legal and Contractual Analysis (V&G Page 124 – 125)

The project will comply with all relevant data protection regulations, verifying all software components are properly licensed, and it will be Implemented with features for data consent management, data encryption, user access controls, and data deletion mechanisms to comply with General Data Protection Regulation (GDPR) and similar laws.

Political Analysis (V&G Page 125)

According to a brand-new system, it requires stakeholder support and adaptation. Key considerations include:

- Role Changes: Addressing shifts in responsibilities and potential resistance.
- Adaptation: Embracing changes and effective adaptation to new systems.
- Communication: Clearly communicating the benefits and changes.
- Training: Providing comprehensive training to ease the transition

Schedules, Timeline and Resource Analysis (V&G Page 124)

XABS is scheduled for completion within 6 months, including phases as following: of planning, analysis, design, development, testing, and deployment, with allocated resources for each phase.

1. Planning phase (Month 1): Define project scope and requirements, Identify key stakeholders, form project team, and initial resource allocation.
2. Analysis Phase (Month 1): Conduct detailed requirement analysis, develop initial system design and architecture.
3. Design Phase (Month 2): Create detailed design documents, develop prototypes and user interface designs.
4. Development Phase (Month 3-4): Code the system components, integrate with existing systems.
5. Testing Phase (Month 5): Conduct unit testing, integration testing, and user acceptance testing, and address any issues identified during testing.

6. Deployment Phase and Post-Deployment Phase (Month 6): Monitor system performance and address any post-launch issues and collect user feedback for future improvements.

Management Issues (V&G Page 129 – 130)

Team Configuration and Management

Project: Xtreme automated booking system		Prepare by: Chawaphong Khunsoongnoen			Legend: P = Primary S = Support		
Manager: Chawaphong Khunsoongnoen		Page: 1 of 1					
		Responsibility Matrix					
Task ID	Task	Ethan	Samy	Nancy	Prim	Lola	Chawaphong
A	Collect requirements		S			S	P
B	Develop automated system	S		S	P		
C	Develop system interface	P			S		S
D	Build database	S		P	S		S
E	Design test scenario	S	P	S	S	S	S
F	Run test scenario	S	S	S	S	S	P
G	System documentation				S	P	S
H	Feedback gathering		S			P	S

Communication Plan

Stakeholder	Document	Format	Team Contact	Date Due
Team members	Project status report	Project intranet	Chawaphong and Samy	First Monday of month
Project sponsor	Project status report	Email/PDF	Chawaphong	First Monday of month
Development team	Technical implementation report	Email/Presentation	Ethan and Prim	Bimonthly
Marketing team	User feedback summary	Document/Presentation	Lola	1 month after deployment day

Project Standards and Procedures

XABS will adopt Agile methodology to allow continuous improvement and flexibility due to rapid changes in the project life cycle to enhance user experience, also focusing on stakeholder collaboration and communication, security and risk management to achieve a comprehensive automated booking system.

Other Project-Specific Topics

- Comprehensive training: staff may face challenges in understanding and using the new functionalities. So, providing a training session for staff to learn about new functionalities and get familiar with the system, also continuous support for any questions or concerns.
- Transition planning: Integrating the new system into daily workflows can cause disruptions and resistance. Gradually integrate a new system into daily workflows to minimize disruption and facilitate smooth adoption to ensure constant communication and involvement of staff throughout the transition.
- Feedback gathering and continued improvement: gather feedback from both users and staff to ensure the system evolves to better meet user needs and enhance the overall experience.

Project Scope Statement

General Project Information

Project Name: Xtreme automated booking system (XABS)

Sponsor: J.C.

Project Manager: Chawaphong Khunsoongnoen

Problem/Opportunity Statement:

The current system of Xtreme Adventure Tours is unreliable and inefficient due to manual processes for managing tour bookings and checking availability. This leads to errors, uncertainties, and reduced customer satisfaction. By automating the booking system, Xtreme Adventure Tours has the opportunity to enhance operational efficiency, provide a seamless booking experience, and improve customer service, ultimately increasing customer confidence and business growth.

Project Objectives:

- Develop a comprehensive automated booking system that consists of unique features including user-friendly booking interfaces, secure payment processing, automated confirmations, and integration with existing customer management systems.
- Develop a real-time availability checking system to Ensure up-to-date tour availability, reduce overbooking, and provide instant booking confirmations.
- The new enhancement systems must have scalability to handle increased demand and the company's growth.

Project Description:

The Xtreme automated system project aims to replace the existing system that requires manual processes to manage tour booking and check availability with a robust automated software solution.

- User Interface: User-friendly interfaces for customers and staff.
- Real-Time Updates: Real-time availability checks and instant booking confirmations.
- Integration: Seamless integration with current systems and database.
- Security: Secure payment processing and data protection.

Key stakeholders:

- Customers: End-users who will be using the new system.
- Staff: Employees responsible for managing bookings and customer interactions.
- Tour company manager: personnel who is in charge of contacting each tour company.
- IT Support Team: technicians who will manage and maintain the system's infrastructure and security.

Business Benefits:

- Enhance Customer Experience and Satisfaction: Expected to increase customer satisfaction by 20% through a smoother booking process and reduced errors.
- Improve Operational Efficiency: Anticipated to reduce booking management time and error by 30%.

- Support Business Growth: Scalable system designed to handle a 50% increase in booking volume over the next two years.

Project Deliverables:

- An automated booking management system: Complete system with all defined features and functionalities.
- A system documentation and manual: Detailed user guides, system architecture documentation, and operational procedures.
- A training session for the new system: Comprehensive training sessions for staff, including hands-on workshops and online resources.
- Maintenance and technical support: Ongoing support services, including regular updates, troubleshooting, and technical assistance.

Estimated Project Duration:

6 months

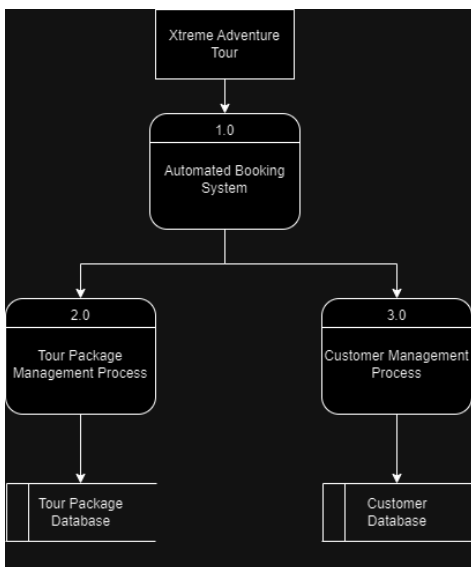
- Month 1: Planning and Requirement Analysis
- Month 2: System Design and Architecture
- Month 3-4: Development and Integration
- Month 5: Testing and Quality Assurance
- Month 6: Deployment, Training, and Post-Deployment

Process Requirements and Data Modeling

Process Requirements (V&G Page 170 – 173)

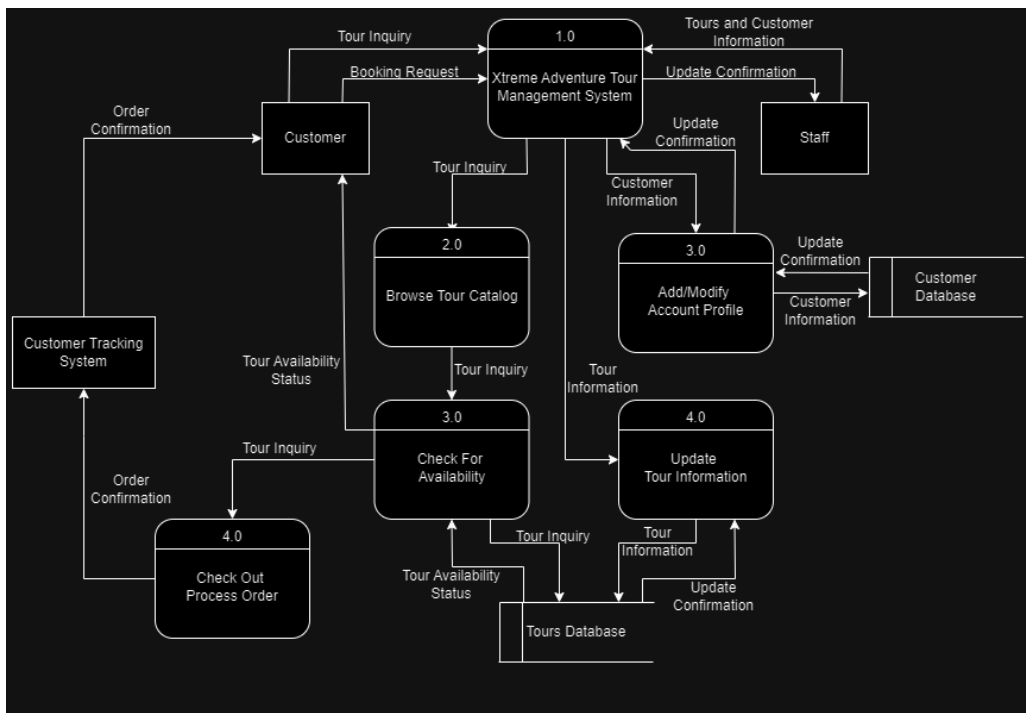
Organization and User Needs

- Automated processes from the new system to reduce errors from manual processes.
- Real-time availability checking system to keep information up to date.
- User-friendly interfaces to enhance user experience including customers and staff.
- XABS must have high scalability to handle the company's growth in future.
- XABS must have high security to earn customers' trust for booking and payment processing.



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177 **Level-0 Data Flow Diagram (DFD)**

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179 Visual of Your DFD



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182 **Description of the DFD**

- 183 - The Xtreme Adventure Tour Management System is the center of other sub-modules that will manage
- 184 booking requests from customers, update real-time tour information, and communication within the system
- 185 and entities.
- 186 - Browse Tour Catalog process allows customers to view tour information by sending tour inquiries.
- 187 - Check For Availability process then send tour inquiries to Tours Database to check for availability then
- 188 send the status back and pass on to the Checkout Process Order if the selected tour has availability.

- The Customer Tracking System then receives order information and finalizes the booking and sends order confirmation to customers.

- Staff can create, update, or modify profiles through the central system by sending information to Customer Database.

- Staff can also manually update the Tour information then the Update process will contact Tour Database then send back the confirmation to staff.

Additions to the DFD

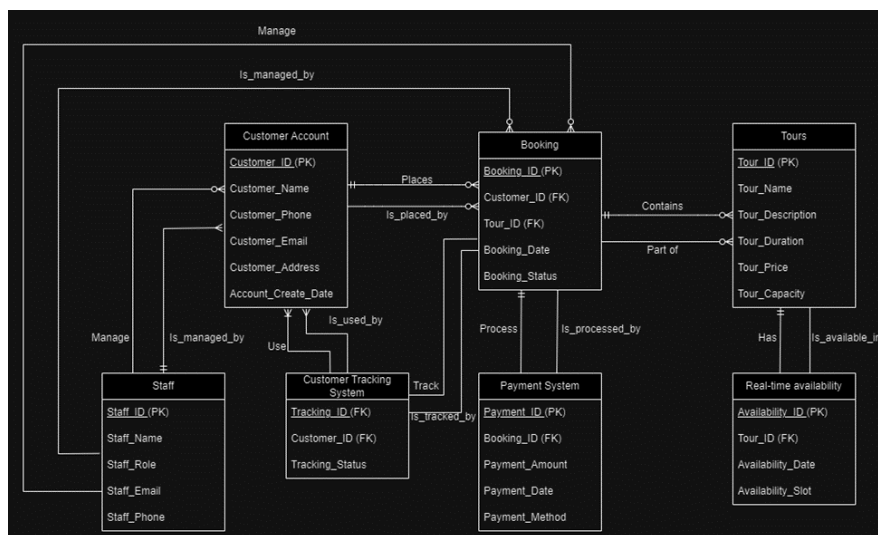
- Customer Feedback Collection Process to allow customers to provide feedback after their tours, then save the feedback in a database to review and analyze them for improvement.

- Payment processing process captures the payment request, verifies the payment details, interact with different payment gateways, then send the payment confirmation.

System Design

Database Schema (V&G Page 298 – 333)

Structure of the Database



Description of the Database Relationships

The structure of Xtreme Automated Booking System (XABS) database to supports all automated functionalities, it includes key entities such as Customer Account, Staff, Booking, Tours, Real-time Availability, Payment System, and Customer Tracking System, along with their attributes and relationships. This structure also ensures data integrity, efficiency, and scalability of the system.

Reports and Forms (V&G Page 341 – 361)

Example Forms and Description of Purpose

- Booking report: The booking report provides a detailed overview of all bookings made within a specific period. The purpose of this booking report is to help in tracking all bookings, monitoring booking trends, managing tour capacities, and ensuring timely follow-up on pending payments. The report includes

- Booking ID: Primary key to identify each booking.
- Customer Name: Name of the customer who made the booking.
- Tour Name: Name of the tour booked.

- Booking Date: Date when the booking was made.
- Booking Status: Current status of the booking.
- Payment Status: Indicates whether the payment has been completed or is pending.

Booking ID	Customer Name	Tour Name	Booking Date	Booking Status	Payment Status
B12345	John Doe	Rafting	2024-06-15	Confirmed	Paid
B12346	Jane Smith	Scuba Diving	2024-06-20	Pending	Unpaid
B12347	Alice Johnson	Kayaking	2024-07-01	Confirmed	Paid
B12348	Bob Brown	Zip-Lining	2024-07-10	Cancelled	Refunded

- Booking form: The booking form collects necessary information from the customer to book a tour. The booking form's purpose is to ensure that all relevant information is collected from the customer to facilitate a smooth booking process to help in properly managing within the system, gather special requests and customize tours for each customer. It includes fields as follows:

- Customer Name: Full name of the customer.
- Email: Email address of the customer.
- Phone number: Phone number of the customer.
- Tour Selection: Dropdown or selection box to choose the desired tour.
- Booking Date: Date when the customer wants to book the tour.
- Number of Participants: Number of people included in the booking.
- Special Requests: Any specific requirements or requests from the customer (e.g., dietary restrictions, accessibility needs).

Booking Form

Customer name :

Email :

Phone number :

Tour selection :

Booking date :

Number of participant:

Special request :

Submit

- Payment form: The payment form provides multiple payment methods, facilitates seamless payment processing, and collects all necessary information from the customer to process their payment securely. The purpose of this form is to ensure that the transaction is completed smoothly and securely, capturing essential payment details and billing information. The form includes

- Payment Method: Selection buttons for choosing the payment method.
- Card Details: Card number, expiration date, and CVV code for credit card payments.
- Billing address: Billing address of the customer

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Payment Form

☐ Credit/Debit Card ☐ Google Pay ☐ Paypal

Card number :

Cardholder name :

Expiration date :

CVV :

Billing address :

Name

Address 1

Address 2

City State Country

Postal code

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250 User Interfaces (V&G Page 368 – 398)

251 User Interface Example

252 - Landing page: The main entry point for users who visit the Xtreme Adventures Tours website from
253 their laptops, also responsive across different platforms such as mobile phone, tablet, etc. This page has a
254 hero section to enhance user experience and has a navigation bar to lead users to every key section of this
255 site. This page also highlights some featured tours, tour recommendation, promotion, and a quick link to some
256 popular activities.

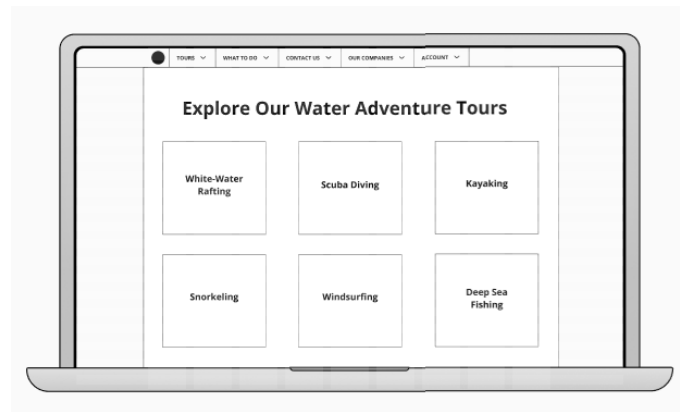


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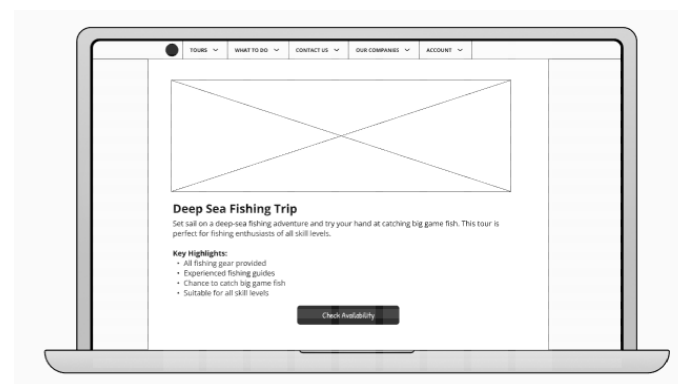
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259 - Adventure browsing page: After the user selected the type of their own interested tour from drop
260 down menu, will lead them to adventure catalog page to allow them to see every tour that we provide. Each
261 tour is displayed with an activity image, title, and a link for more information. This page will also have a filter
262 function to specify their own preferences, also to see availability.

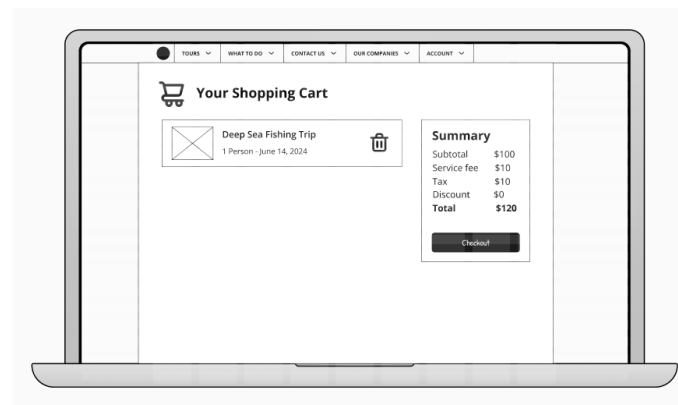
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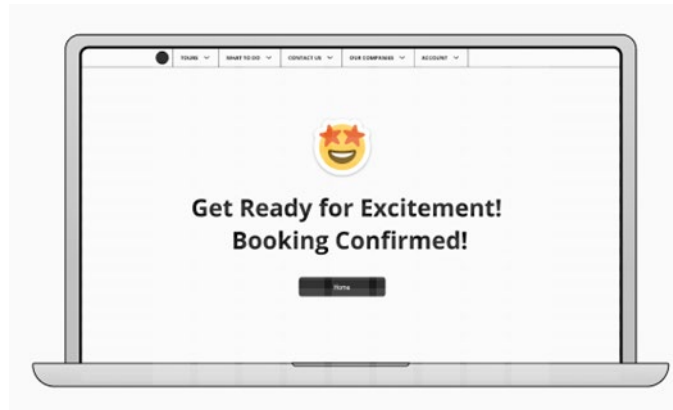
- Tour detail page: The next page after the user selects the tour will provide comprehensive information about the specific tour including high-quality images or video, description, tour itineraries, pricing, availability, and customer reviews.



- Checkout page: The checkout page is where users finalize their own bookings. This page will display the tours that they chose, price breakdowns, some suggestions, and allow them to modify their order. This page will also provide different payment methods to allow users to pay in various reliable secure methods.



- Confirmation page: The confirmation page will notify users that their bookings are successful, and send the confirmation email which includes order number, order detail, tracking link, etc. to their email address.



Factoring Usability, Accessibility and User Experience

The Xtreme Adventure Tour Management System (XABS) will adopt cloud computing to provides robust solutions for enhancing scalability, security, and maintainability to handle future growth and increased usage such as Amazon Web Services (AWS), Microsoft Azure, etc.

Addressing Scalability

- AWS Auto Scaling: Use Auto Scaling to automatically adjust EC2 instances based on demand to be able to handle growth number of users without performance degradation.*
- AWS Elastic Load Balancing (ELB): Distribute incoming traffic across multiple instances to avoid any single server getting overloaded, ensuring your system remains fast and reliable.*

Addressing Security

- AWS Identity and Access Management (IAM): Set up IAM to control who can access your AWS resources by giving people only the permissions depending on their roles to keep the system secure and reduce the risk of unauthorized access.*
- AWS Key Management Service (KMS): Use KMS to encrypt data to ensures that someone without the decryption keys gains access and read data.*

Addressing Maintainability

- AWS CloudFormation: Adopt CloudFormation to manage your infrastructure as code. This makes it easy to deploy, update, and maintain your resources, ensuring everything is consistent and version controlled.*
- Azure Monitor: Utilize Azure Monitor to monitor the system's performance and health to help catching and addressing issues before they become more serious issues, ensuring smooth operation.*
- Comprehensive Documentation: Provide a comprehensive documentation of XABS's architecture, configurations, and procedures to help everyone is on the same page and onboarding new team members much easier.*

Prototyping (V&G Page 368 – 398)

Example Prototype and Description of Function

The prototype demonstrates the user experience through each screen of the new system, including the landing page, adventure browsing page, tour detail page, checkout page, and confirmation page. It illustrates a happy path scenario where the user searches for the deep-sea fishing tour and decides to book it through the new system.

Prototype link: <https://www.figma.com/proto/PBEKUkm7hFUCHWsLuN9Ca0/IS421-Xtreme-Adventures-Tours-Prototype?page-id=0%3A1&node-id=2-494&viewport=-149%2C337%2C0.06&t=Isi7B0WAIUDL8rw-1&scaling=min-zoom&starting-point-node-id=2%3A494>

Implementation and Maintenance

System Installation Plan (V&G Page 450 - 453)

Type of Installation: Phased installation

Required Resources:

- *Hardware: Cloud infrastructure (AWS EC2 instances, S3 storage)*
- *Software: Xtreme Automated Booking System (XABS), Database management, AWS services*

Timeline:

- *Month 1: Project planning, Preparation, Installation, Resource allocation*
 - *Tasks: define project scope and objectives, allocate resources and form project team, set up initial infrastructure (cloud instances, storage).*
 - *Personnel Involved: Project manager, system architects, IT support team.*
 - *Contingency Plan: If delays or issues occur, extend the planning phase by two weeks to ensure proper setup.*
- *Month 2: Customer Account Management installation and testing, provide user training*
 - *Tasks: Install and configure customer account management module, perform initial testing, conduct training sessions for staff.*
 - *Personnel Involved: Project manager, developers, QA testers, staff, training instructor.*
 - *Contingency Plan: Allocate extra resources for testing if initial tests fail, and schedule additional training sessions if needed.*
- *Month 3: XABS installation and testing, Provide user training*
 - *Tasks: Install and configure the main booking system (XABS), perform testing, conduct training sessions.*
 - *Personnel Involved: Project manager, developers, QA testers, staff, training instructor.*
 - *Contingency Plan: Allocate extra resources for testing if initial tests fail, and schedule additional training sessions if needed.*
- *Month 4: Automated payment system installation and testing, Provide user training*
 - *Tasks: Install and configure the payment processing module, perform comprehensive testing to ensure secure transactions, provide training sessions*
 - *Personnel Involved: Project manager, developers, QA testers, training instructor.*
 - *Contingency Plan: Engage external consultants if there are issues with payment integration and provide additional training sessions as required.*
- *Month 5: All modules integration, full system testing*
 - *Tasks: Integrate all modules into a cohesive system, conduct a full system testing to ensure all components work together properly and seamlessly.*
 - *Personnel Involved: Project manager, developers, QA testers.*

- *Contingency Plan: Allocate additional time for debugging and re-testing if integration issues arise.*
- *Month 6: Deployment and Feedback gathering*
 - *Tasks: Deploy the system to production environment, monitor system performance and gather feedback from users, make any necessary adjustments based on user feedback.*
 - *Personnel Involved: Project manager, IT support team, staff.*
 - *Contingency Plan: Have a backup plan in case critical issues arise and schedule follow-up sessions to address feedback.*

Risk Identification and Mitigation (V&G Page 458 - 463)

System Compatibility

Issue or Risk	Mitigation Strategy
Integration with existing systems	Gradually integrate each module with extensive compatibility testing. Provide comprehensive user training and promptly resolve any issues that occur during integration.
Operational cost overflow	Develop a detailed budget plan including all potential costs associated with integration. Monitor expenditures closely and adjust the plan as needed. Implement cost-saving measures such as optimizing resource usage.

Data Migration

Issue or Risk	Mitigation Strategy
Data loss	Use robust data migration tools and perform multiple tests to ensure data integrity before the actual migration. Implement a comprehensive backup and recovery strategy to restore data in case of any loss.
Data corruption	Validate data integrity during the migration process with checksums and validation scripts. Maintain backup copies of all data and perform regular integrity checks throughout the migration process.

User Training

Issue or Risk	Mitigation Strategy
Incomprehensive training programs	Develop comprehensive training programs including workshops, hands-on sessions, and detailed user manuals. Tailor training sessions to different user roles to ensure all aspects of the system are covered adequately.
Users' resistance to training	Provide flexible training options such as online tutorials, webinars, and in-person sessions to cater to different learning preferences. Develop engagement programs to emphasize the importance and benefits of the new system, and involve key users in the training development process to increase acceptance.

System Testing

Issue or Risk	Mitigation Strategy
Undetected bugs	Implement rigid testing protocols including unit testing, integration testing, and user acceptance testing. Ensure comprehensive test coverage for all system functionalities and involve end-users in the testing process to identify usability issues
Hardware failures	Use redundant hardware setups to ensure high availability. Implement regular maintenance schedules and monitoring systems to detect issues early. Have a contingency plan in place for quick replacement or repair of faulty hardware.

Security Threats

Issue or Risk	Mitigation Strategy
Cyber attack	Implement strong encryption, access controls, and regular security audits. Conduct penetration testing to identify and address vulnerabilities. Keep security systems and protocols up-to-date with the latest patches and updates..

System Maintenance and Upgrade (V&G Page 472 – 485)

Post-Implementation Maintenance Plan:

- Corrective Maintenance:

- Design repairing and programming errors debugging (Regularly monitoring the system, and keep track and resolve issues promptly)
- Schedule: Weekly system checks and real-time monitoring.
- Responsibility: IT support team and developers.

- Adaptive Maintenance:

- Modify the system to accommodate environmental changes (Make the system stay up-to-date with new technologies, user requirements, etc.)
- Schedule: Quarterly reviews to assess the need for updates and modifications.
- Responsibility: System architects and developers.

- Perfective Maintenance: Collect user feedback, analyze, identify areas of improvement

- Collect and analyze user feedback to identify areas of improvement. Implement changes to enhance system performance and user satisfaction.
- Schedule: Monthly feedback collection through surveys and user feedback sessions.
- Responsibility: Business analysts and user experience (UX) designers.

- Preventive Maintenance:

- Implement automated monitoring and alert system, conduct regular security audits, scheduling regular updates, data backup and recovery plan.
- Schedule: Monthly security audits and updates, daily automated backups.
- Responsibility: IT security team and database administrators.

414 *Strategy for System Upgrades:*

415 *- Technology enhancements:*

- 416 ○ *Keep the system up-to-date with the latest technology, regularly check system's healthiness,*
- 417 *improve system's performance.*
- 418 ○ *Schedule: Semi-annual technology assessments*
- 419 ○ *Responsibility: IT support team and system architects*

420 *- User-involving modifications:*

- 421 ○ *Keep iterates through user feedback survey sessions and adjust design to keep improving user*
- 422 *experience.*
- 423 ○ *Schedule: Bi-monthly user feedback sessions and iterative design updates*
- 424 ○ *Responsibility: UX designers and business analysts*

425 *- Scalability improvements:*

- 426 ○ *Evaluate and enhance the system scalability both horizontally and vertically, implementing*
- 427 *auto-scaling to handle growing amounts of users.*
- 428 ○ *Schedule: Annual scalability assessments and as-needed adjustments.*
- 429 ○ *Responsibility: System architects and cloud infrastructure managers.*

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